Essays on Causal Mechanism and Causal Generalization

by

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Dedication

To all who have helped me.

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Abstract

The credibility revolution advances the use of research designs that permit identification and estimation of causal effects. However, understanding which mechanisms produce measured causal effects remains a challenge. A dominant current approach to the quantitative evaluation of mechanisms relies on the detection of heterogeneous treatment effects with respect to pre-treatment covariates. Chapter 1 develops a framework to understand when the existence of such heterogeneous treatment effects can support inferences about the activation of a mechanism. We show first that this design cannot provide evidence of mechanism activation without additional, generally implicit, assumptions. Further, even when these assumptions are satisfied, if a measured outcome is produced by a non-linear transformation of a directly-affected outcome of theoretical interest, heterogeneous treatment effects are not informative of mechanism activation. We provide novel guidance for interpretation and research design in light of these findings.

Understanding causal mechanisms is essential for explaining and generalizing empirical phenomena. Causal mediation analysis offers statistical techniques to quantify mediation effects. However, existing methods typically require strong identification assumptions or sophisticated research designs. In Chapter 2, we develop a novel identification strategy that simplifies these assumptions, enabling the simultaneous estimation of causal and mediation effects. The strategy is based on a new decomposition of total treatment effects, which transforms the challenging mediation problem into a simple linear regression problem. We demonstrate that the primary source of identification power resides in the heterogeneous treatment effects on the mediator. To

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illustrate the efficacy of our method, we apply it to estimate the causal mediation effects in two studies, focusing on common pool resource governance and voting information. Furthermore, we have developed statistical software to facilitate the implementation of our method.

Can causal effects estimated in experiment be generalized to real-world scenarios? This question lies at the heart of social science studies, where the ultimate concern is the real-life impact of research findings. External validity primarily assesses whether experimental effects persist across different settings, including populations, treatments, outcomes, and contexts, implicitly presuming the experiment's ecological validity—that is, the consistency of experimental effects with their real-life counterparts even without dramatic varying those settings. However, we argue that this presumed consistency may not always hold, especially in experiments involving multi-dimensional decision processes, such as conjoint survey experiments. In Chapter 3, we introduce a formal model to elucidate how attention and salience effects lead to three types of inconsistencies between experimental findings and real-world phenomena: amplified effect magnitude, effect sign reversal, and effect relative importance reversal. We derive testable hypotheses from each theoretical outcome and test these hypotheses using data from various existing conjoint experiments. Drawing on our theoretical framework, we propose several guidelines for experimental design aimed at enhancing the generalizability of survey experiment findings.

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1 Introduction

Causal Identification revolution has spurred widespread adoption of research designs aimed at facilitating the unbiased estimation of causal effects. Yet, these designs often fall short of elucidating the mechanisms—the why and how—behind the observed treatment effects on outcomes. The quest to uncover these underlying mechanisms is pivotal for leveraging empirical evidence to deepen our understanding of social phenomena.

A significant challenge in current methodologies is their reliance on stringent assumptions that are difficult to validate in practical settings. Additionally, certain methods prevalent in contemporary research have yet to undergo thorough scrutiny. In my dissertation, I introduce several straightforward methodologies for estimating both causal mediation effects and causal effects, applicable to observational studies and experiments alike. Through formal theory, I elucidate the underlying logic behind these challenges, contributing to a more nuanced understanding of causal inference in social science research.

In the first chapter, joint with Tara Slough, we explore the idea of deriving causal mechanisms from heterogeneous treatment effects (HTEs). We ask: when can heterogenous treatment effects provide evidence about mechanism activation? In applied studies, researchers often use HTEs to evaluate or test mechanisms. A review of the 2021 volumes of three premier political science journals reveals that over half of the studies report HTE, with 87% employing HTE as a means to detect mechanisms. Despite its widespread application, the theoretical foundation is not well explored. Therefore, we construct a framework and demonstrate that some conclusions can be

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misleading. First, HTEs do not provide evidence of mechanism activation without additional exclusion assumptions. Second, we find that the relationship between HTEs and mechanisms is contingent upon the theoretical nature of the outcome variable. Some outcomes are indirectly affected by a mechanism. When measured outcome is produced by a non-linear transformation of a directly-affected outcome of theoretical interest, HTEs are not informative of mechanism activation. Our framework also generates concrete guidelines for improving the use of HTEs for mechanism testing.

In the second chapter, I propose a novel identification strategy to non-parametrically identify causal mediation effects. Existing methods typically require sequential ignorability assumptions which hold that both the treatment and the observed mediator are ignorable. My identification strategy innovatively circumvents these assumptions. The strategy is based on a novel decomposition of total treatment effects, which transforms the challenging mediation problem into a simple linear regression problem without compromising the non-parametric nature. My method promises the consistent estimation of the causal mediation effect under weak assumptions. Additionally, I have created statistical software combining machine learning techniques to facilitate the implementation.

The last chapter studies a fundamental generalization problem of the multi-dimensional experiment. Can the internally valid effects observed in experiment be generalized to real-world scenarios? This question lies at the heart of social science studies, where the ultimate concern is the real-life impact of research findings. External validity primarily assesses whether experimental effects persist across different settings, including populations, treatments, outcomes, and contexts, implicitly presuming the experiment's ecological validity—that is, the consistency of experimental effects with their real-life counterparts even without dramatic varying those settings. However, we argue that this presumed consistency may not always hold, especially in experiments involving multi-dimensional decision processes, such as conjoint survey experiments. We introduce a formal model to elucidate how attention and salience effects lead to three types of

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inconsistencies between experimental findings and real-world phenomena: amplified effect magnitude, effect sign reversal, and effect relative importance reversal. We derive testable hypotheses from each theoretical outcome and verify these hypotheses using data from various existing conjoint experiments, in addition to conducting our own experiments. Drawing on our theoretical framework, we propose several guidelines for experimental design aimed at enhancing the generalizability of survey experiment findings.

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2 | Heterogeneous Treatment Effects and Causal Mechanisms

ABSTRACT

The credibility revolution advances the use of research designs that permit identification and estimation of causal effects. However, understanding which mechanisms produce measured causal effects remains a challenge. A dominant current approach to the quantitative evaluation of mechanisms relies on the detection of heterogeneous treatment effects with respect to pre-treatment covariates. This paper develops a framework to understand when the existence of such heterogeneous treatment effects can support inferences about the activation of a mechanism. We show first that this design cannot provide evidence of mechanism activation without additional, generally implicit, assumptions. Further, even when these assumptions are satisfied, if a measured outcome is produced by a non-linear transformation of a directly-affected outcome of theoretical interest, heterogeneous treatment effects are not informative of mechanism activation. We provide novel guidance for interpretation and research design in light of these findings.

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2.1 Introduction

The credibility revolution in empirical social science has motivated the largescale adoption of research designs that facilitate unbiased estimation of causal effects [12, 113]. Such unbiased estimates allow for valid measurement of the effect of a treatment on an outcome. However, they do not generally provide evidence about *why* or *how* the treatment affected the outcome. These questions of why or how are ultimately questions about the activation and influence of causal mechanisms. Understanding the mechanisms that produce causal effects is central to our ability to use empirical evidence to understand social phenomena [119].

Applied researchers typically pursue a number of different approaches to ascertaining the mechanisms that generate causal effects. There exist at least four distinct approaches in the applied literature: (1) evaluation of the *sign* of treatment effects on a given outcome [e.g., 15]; (2) mediation analysis [60, 78, 80, 81]; (3) multimethod research involving some form of qualitative or quantitative triangulation of causal findings [45, 92]; and (4) the estimation of heterogeneous treatment effects (HTEs). The last approach—HTEs—compares estimated treatment effects for various subgroups thought to be informative about mechanism activation. While this is currently the modal approach to (quantitative) mechanism-testing in political science, the theoretical properties of this enterprise are not well explored.

To examine the prevalence of the use of HTEs to learn about causal mechanisms, we survey the 2021 volumes of three leading journals in political science: the *American Journal of Political Science* (AJPS), the *American Political Science Review* (APSR), and the *Journal of Politics* (JoP). We first identify the subset of papers that analyze quantitative empirical data. We then report the proportion of those empirical papers that report HTEs or subgroup-specific treatment effects. Finally, we report the proportion of papers using HTEs that interpret these quantities as providing information about causal mechanisms(s). Table 2.1 shows that in each of the three leading journals, a majority of quantitative studies estimate HTEs. Moreover, conditional on reporting

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Journal (Volume)	Number of articles: Total	Quant. empirical	Reporting HTEs	Pr(Report HTEs | Quant. empirical)	Pr(Mechanism test | Report HTE)
AJPS (65)	61	41	24	0.59	0.83
APSR (115)	102	75	42	0.56	0.90
JoP (83)	142	106	59	0.56	0.76
Total	305	222	125	0.56	0.82

Table 2.1: Classification
Authors' classification of articles published in three leading political science journals in 2021.

any HTEs, the vast majority of articles (82% across three journals) interpret these quantities as providing information about mechanisms. Collectively, these figures indicate that almost half (46%) of recent quantitative empirical articles in these journals use HTEs to assess mechanisms. Table 2.9.1 further documents that the share of studies that use of HTE for mechanism detection is similar across all quantitative research designs/identification strategies in common usage.

Existing concerns about HTEs have largely focused on the *statistical* properties of relevant estimators and hypothesis tests. In particular, interaction effects are known to have low statistical power [e.g., 99]. Moreover, estimation of HTEs with respect to many (pre-treatment) covariates risks multiple comparisons problems [53, 59, 90]. While these criticisms are important, they are distinct from *theoretical* questions about how the presence or absence of HTEs links to causal mechanisms. We take on this challenge by asking: under what conditions do HTEs provide evidence of mechanism activation?

To answer this question, we develop a framework to formally link HTEs with respect to a covariate to the effect of a specific mechanism. To do so, we extend the workhorse causal mediation framework [78]. Within our framework, a *mechanism* is an underlying process that influences experience in order to produce a (causal) effect if it is activated [119]. A *mediator*, or mechanism representation, should thus be affected by treatment and have a non-zero (indirect) effect on the outcome if the mechanism is active for some unit. In order to use HTEs to detect mechanisms, empiricists rely on a measured *moderator* or pre-treatment covariate that is thought to predict the degree to which treatment activates a mechanism and/or the mechanism's effect on an outcome

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of interest. Our results characterize the conditions under which heterogeneity in conditional average treatment effects (CATEs) with respect to a given moderator is a sufficient condition to show that the mechanism is active. A mechanism is *active* when its indirect effect is non-zero for some unit.

We first present two classes of exclusion assumptions that link HTEs to a specified mechanism. These assumptions hold that the moderator of interest is excluded with respect to (1) the average indirect effect of other mechanisms and (2) the average direct effect of the treatment on the outcome. The intuition for these assumptions is straightforward. The use of HTEs to assess the activation of a given mechanism relies on the additive separability of the effect of that mechanism from the effects of other mechanisms. These exclusion assumptions are needed to ensure that additive separability is possible. In their absence, the relationship between HTEs and the indirect effect of a mechanism is unspecified. In this sense, these assumptions are implicitly invoked by current practice, but are not explicitly stated or defended.

When these exclusion assumptions hold, what can we learn about a mechanism of interest from HTEs with respect to a given covariate? Our main results characterize what we can learn from the existence or non-existence of HTEs. We show that whether such learning is possible depends on the relationship between measured outcomes and the theory. We thus distinguish between outcome variables that are theorized to be *directly affected* by a mechanism versus (causally) subsequent outcomes that are *indirectly* affected.

First, consider the case in which a measured outcome is directly affected by the mechanism of interest. We show that when both exclusion assumptions hold, the existence of HTEs provides evidence of mechanism activation. This broadly conforms to current practices in applied research, albeit while making the underlying assumptions explicit. However, when there do not exist HTEs, there are two possible explanations: (1) the indirect effect of a mechanism is not moderated by any covariate; and/or (2) the theorized relationship between the covariate and the indirect effect of the mechanism was misspecified. Neither possibility distinguishes between an active or inactive

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mechanism. This means that a lack of HTEs cannot "rule out" a mechanism by showing that it is inactive. This finding contradicts standard interpretations of (the absence of) HTEs in the applied literature.

We further consider the common case in which we observe outcomes that are *indirectly* affected by a mechanism. For example, in a political economy model, a mechanism may change an actor's utility, but researchers measure only the actor's discrete choice/behavior. In political psychology, a treatment may affect a subject's (latent) attitudes, but researchers measure their survey response on a Likert scale. When an outcome is generated by a non-linear mapping of the directly-affected outcome (as in both of these examples), the relationship between HTEs and mechanisms changes. In this case, the existence of HTEs with respect to a covariate no longer provides evidence of mechanism activation, even when both exclusion assumptions hold. The logic for this result is straightforward: the non-linear mapping of the mechanism's influence into an observed outcome breaks the additive separability of the indirect effect of interest from other indirect and direct effects, which undermines our ability to link HTEs to the indirect effect of a mechanism.

This paper makes three principal contributions. First, we contribute to literature on varying uses of HTEs. Our focus, in line with most current empirical applications, is on the use of HTEs to learn about the mechanisms that produce treatment effects. It is crucial to distinguish this of HTEs from the use of HTEs for extrapolation, prediction, or targeting of treatment. The latter class of concerns—extrapolation, prediction, and targeting—that are more dominant in the recent methodological literature [18, 41, 48, 87]. Whereas these applications are enhanced by the use of machine learning methods for the detection of heterogeneity [e.g., 17, 61], we show that using HTEs to learn about mechanisms instead relies on a deductive theoretical mapping of covariates to mechanisms, which is unlikely to be advanced by developments in machine learning.

Second, we expand a growing literature on the theoretical implications of empirical models (TIEM) [2, 14, 15, 29, 116]. We make two central interventions to this literature. First, our

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framework makes explicit links between a causal mediation framework that is used more prominently by empiricists and (formal) theoretical models. Second, we introduce questions about how measured outcomes relate to theoretical constructs by distinguishing between directly- and indirectly-affected outcomes. While measurement is central to recent TIEM work on evidence accumulation in a cross-study environment [119–121], it has not been widely explored in the single-study environment.

Finally, we provide practical guidance for empirical researchers who want to learn about which mechanisms generate observed effects. Our assumptions and results reveal a minimal set of attributes of an applied theory that can support the use of HTEs to learn about mechanisms. Two of these attributes, the relationships between (1) a covariate of interest and other mechanisms and (2) measured outcomes and theoretical objects of interest, are generally not discussed in applied work. Second, we show how interpretation of HTEs can be improved, returning to the statistical problems that are well known in this literature. Third, we discuss how our analysis can be used to inform prospective research design. Last, we show that in order to infer mechanism activation from HTEs in the case of indirectly-observed outcomes, stronger theoretical assumptions about the mapping from directly-observed to indirectly-observed outcomes are necessary. Collectively, these suggestions allow practitioners to accurately use—or, when indicated, avoid—HTEs as a quantitative test of mechanisms.

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2.2 Current Practice

As reported in Table 2.1, 82% of the articles that report HTEs interpret these quantities as tests of a mechanism. Two interpretations of HTEs are common. First, the *presence* of HTEs with respect to a specific covariate provides evidence that a mechanism is active. For example, Haim, Ravanilla, and Sexton [64] report the results of a field experiment in a conflict-affect region of the Philippines that randomized provision of a program that connected villages to state services and

sought to increase village leaders' trust in the state. In the context of the COVID-19 pandemic, they show that the program increased village leaders' probability of reporting COVID risk information to a government task force by $\approx$ 10 percentage points. Haim, Ravanilla, and Sexton [64] argue that the program increased reporting by increasing beliefs in government competence. They report HTEs that indicate the effect on reporting was significantly larger in communities in which village leaders initially believed that the government "[did] not have capacity to meet needs." Here, the presence of HTEs with respect to perceived capacity is thought to provide evidence that leaders assigned to treatment updated their beliefs about government capacity (the mechanism), thereby increasing their willingness to report risk information to the government.

Second, the *absence* of HTEs with respect to a given covariate is frequently used to "rule out" the activation of a possible mechanism (often called an alternative explanation). For example, Moscowitz [100] argues that the proportion of in-state residents in one's local media market increases rates of voter political knowledge and split-ticket voting in the US, countering trends toward the nationalization of politics. The paper provides evidence that news coverage is the mechanism that drives this effect. However, it also seeks to rule out an alternative mechanism: campaign advertising. The logic for this alternative mechanism that holds that more in-state residents leads to more campaign advertising about in-state candidates, which in turn increases voter knowledge. Since advertisements air when during election season (when incumbents contest reelection), Moscowitz [100] codes an indicator that takes the value of "1" when the survey was fielded during an election season and "0" otherwise. The effect of the share of in-state residents in a local media market does not change during election seasons. This lack of heterogeneity is used to provide evidence against the advertisement-based mechanism.

Both of the above examples are exceptionally clear in delineating the mechanisms under consideration using HTEs, and thereby serve as exemplars of current practice. We show how current practice with respect to HTEs and the detection of causal mechanisms can mislead. We proceed from these empirical examples to a purely theoretical motivating example to illustrate the con-

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cern.

2.3 Motivating Example: Exogenous Shocks and Voting

Consider a large class of natural experiments on the effect of some exogenous shock, denoted ω, on voter beliefs and behavior in a democracy.[1] This shock could be a natural disaster [e.g. 6, 71], a disease outbreak [e.g., 20], an economic crisis [e.g. 136], or even a seemingly-irrelevant event [e.g. 72]. In order to characterize the effect of the shock on voter beliefs and behavior, we adapt a formal model by Ashworth, Bueno de Mesquita, and Friedenberg [16].

We will assume that an incumbent at the time of the shock is of type $\theta \in \{\underline{\theta}, \overline{\theta}\}$, where $\overline{\theta} > \underline{\theta}$, such that a politician of type $\overline{\theta}$ is "good type" and a politician of type $\underline{\theta}$ is a "bad type." Voters do not observe the politician's type directly, but may be able to learn about their type from observed governance outcomes. The governance outcome is given by:

$$g = f(\theta, \omega) + \varepsilon. \tag{2.1}$$

In this formulation, higher values of ω correspond to a more adverse shock (e.g., a more intense natural disaster). Function f is monotonically increasing in θ and decreasing in ω. ε is an idiosyncratic shock to the governance outcome that is drawn from a symmetric, continuously differentiable probability density function, ϕ, that satisfies the monotone likelihood ratio property relative to g.[2]

Each voter's utility from a politician depends on the politician's type, θ, and a valence shock for the incumbent, v_i. Here, valence captures any attribute of the incumbent that does not depend on their type, including but not limited to bias toward a candidate or ideological closeness.

[1] One could alternatively conduct a survey or field experiment in which researchers provide information on an exogenous shock that is otherwise imperfectly observed by voters.

[2] Formally, this implies that if $x' > x$, then $\frac{\phi(g-x')}{\phi(g-x)}$ is strictly increasing in g.

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